

CRIMENET AI — FORENSIC INTELLIGENCE PLATFORM

Autonomous Multi-Sensor Forensic Intelligence & Criminal Syndicate Link Analysis Platform

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Officer Badge: CYBER-INV-2026-09
Clearance: Level 5 Forensic Investigator

Target Domain: Cyber Forensics & Law Enforcement
Deployment: crimenet-ai-two.vercel.app
Backend Microservice: crimenet-ai.onrender.com

Legal Framework: Section 63(4) BSA 2023 / DPDP Act 2023
Benchmark F1-Score: 0.967 (96.7% Precision/Recall)
Active Case: c1 (Operation Blue Thunder)

PART 1 — EXECUTIVE PROJECT INTRODUCTION

- 1. One-Line Definition:** CrimeNet AI is an end-to-end investigative decision-support platform that fuses cellular records (CDR), hawala ledgers, dark-web intercepts, and vehicle surveillance into an interactive knowledge graph to uncover syndicate kingpins and money-laundering loops with cryptographically verifiable electronic evidence ledgers.
- 2. 30-Second Recruiter Pitch:** Modern criminal investigations suffer from data silos: phone logs in Excel, banking in CSVs, and surveillance footage in closed systems. CrimeNet AI automates this ingestion, runs 3-Tier Adaptive Geolocation, uses Sybil-resistant graph PageRank and a Hidden Kingpin Isolation Index to unmask shadow bosses, and auto-generates court-admissible electronic evidence certificates certified under Section 63(4) of India's new Bharatiya Sakshya Adhiniyam (BSA) 2023.
- 3. 1-Minute Technical Pitch:** I built CrimeNet AI to solve the 'silent kingpin' problem. Standard PageRank flags talkative lieutenants or call centers, missing the true kingpin who gives brief commands over burner phones. CrimeNet AI computes a novel Hidden Kingpin Isolation Index $((\text{Betweenness} / (\text{Degree} + 0.04)) * (\text{Risk} / 50))$ with exponential time decay to penalize adversarial call dilution. For multi-sensor positioning, our 3-Tier Geolocation solver uses Weighted Least Squares in dense urban zones (up to +12.4m), falling back to bicell arcs (+185m) or sector azimuth (+850m) in rural outposts. Every evidence artifact is anchored into an immutable SHA-256 binary Merkle tree with statutory BSA 63(4) PDF certification.
- 4. 2-Minute Comprehensive Explanation:** CrimeNet AI addresses four foundational bottlenecks in modern cyber forensics: (1) Pre-ingestion file tampering is prevented via Telecommunication Service Provider (TSP) manifest validation; (2) Multivariate anomaly vectors (nocturnal call spikes, midnight wires, SIM swaps) are isolated using an Isolation Forest ensemble tuned on 10,000 synthetic records with 96.7% F1-score; (3) Privacy is enforced under the DPDP Act 2023 via automated PII phone masking (+91 98XX-XXX-432), AES-256-GCM envelope encryption, and 30-day intruder webcam retention purge cycles; and (4) Evidence admissibility is guaranteed through automated Section 63(4) BSA 2023 certificates complete with hardware MAC, OS kernel verification, and certifying officer oaths.
- 5. Problem Statement & Real-World Solved:** Police units face catastrophic evidence dismissal in court due to broken chains of custody and weeks of manual Excel correlation. CrimeNet reduces investigation triage time from weeks to minutes while ensuring 100% legal admissibility.
- 6. Target Users:** State Police Cyber Cells (CID), Enforcement Directorate (ED), Financial Intelligence Unit (FIU), Intelligence Bureau (IB), and Special Judicial Magistrates.
- 7. Key Objectives:** Cross-correlate 4 sensor modalities; mathematically unmask stealth kingpins; enforce BSA 2023 Sec 63(4) court admissibility; and maintain strict DPDP Act 2023 privacy governance.
- 8. Main Features:** Cytoscape 48-node syndicate graph, 3-Tier Adaptive Geolocation, Benford's Law hawala audit, Darknet Tor OSINT scanner, SHA-256 Merkle ledger, and ZNCC biometric face authentication with eye-blink liveness.
- 9. Realistic Investigation Scenario (Operation Blue Thunder - Case c1):** Investigating Officer Aditya Pawar ingests cellular CDRs and hawala bank statements. CrimeNet verifies the TSP SHA-256 manifest, detects a midnight wire of Rs 1.5 Cr coupled with a 02:00 AM call burst, trilaterates the burner SIM to Goregaon Industrial Warehouse (+12.4m), activates the Stealth Kingpin Lens to expose Arjun Mehta bridging the hawala channel to Dubai shell entity Phoenix Trading LLC, and exports an official Section 63(4) BSA 2023 certificate ready for judicial submission.

END-TO-END INVESTIGATIVE DATA PIPELINE & EVIDENCE LIFECYCLE



Figure: CrimeNet AI Multi-Sensor Evidence Ingestion & Forensic Processing Pipeline

PART 2 — COMPLETE TECHNOLOGY STACK

Layer / Area	Technology	Exact Purpose in CrimeNet	Location	Simple Explanation
Frontend UI	React 19 & TypeScript	Reactive component architecture, state management, static typing	frontend/src/	Modern web framework ensuring fast, type-safe user interactions.
Bundler	Vite 8.2	Sub-second hot-module replacement and optimized client bundling	frontend/vite.config.ts	Tool that packages and builds frontend code into fast production assets.
Graph Visualization	Cytoscape.js + fcose	Physics-driven force-directed rendering of 48 syndicate suspects & 112 links	GraphExplorer.tsx	Renders interactive node-and-wire visual crime syndicate networks.
GIS Mapping	Mapbox GL 3.29	High-resolution dark-mode map tiles, GPS tracking, and ANPR vehicle breadcrumbs	GeospatialRadar.tsx	Displays interactive satellite maps and moving vehicle coordinate markers.
Charts & Metrics	Recharts 3.10	Anomaly score distribution, ROC-AUC curves, Benford fraud bar charts	Analytics.tsx, ModelEval	Draws clean bar, line, and area charts for forensic data analytics.
Real-Time Comms	Socket.IO Client & Server	Bi-directional event push for real-time raid alerts and radar positions	App.tsx, backend/main.py	Keeps a continuous live connection open between browser and server.
Backend Framework	FastAPI (Python 3.11)	Asynchronous high-performance REST API with automated Pydantic validation	backend/app/main.py	Fast Python web microservice handling all investigative business logic.
Graph Mathematics	NetworkX 3.0	PageRank, betweenness centrality, Louvain community modularity, kingpin index	main.py:840-970	Python library for advanced network graph theory and topology math.
Machine Learning	Scikit-Learn 1.3	Isolation Forest ensemble (250 trees) for multivariate anomaly vector detection	main.py:1100-1280	Machine learning library used for finding hidden outlier transactions.
Database	SQLite3 (8 Tables)	Relational storage of cases, audit logs, alert reviews, chat history, notifications	backend/crimenet.db	Lightweight, reliable SQL database storing all investigative records.
PDF Generation	ReportLab 4.0	Programmatic generation of judicial dossiers and BSA 63(4) statutory certificates	main.py:1880-2180	Compiles official, court-admissible A4 legal PDF documents.
PII Encryption	AES-256-GCM	NIST-compliant envelope encryption with 96-bit nonce for sensitive citizen data	main.py:53-96	Military-grade authenticated encryption keeping personal data secure.
Password Security	PBKDF2-HMAC-SHA256	NIST SP 800-132 password hashing with 100,000 iterations and 16-byte crypt salt	main.py:345-375	Cryptographic key-stretching function protecting investigator passwords.

PART 3 — SYSTEM ARCHITECTURE

CrimeNet AI is architected as a decoupled, zero-trust forensic platform. The client presentation layer communicates with the FastAPI backend gateway via secure Bearer JWT tokens. All requests pass through Role-Based Access Control (RBAC) with 4 hierarchical clearance tiers: Supervisory Officer (4), Lead Investigator (3), Forensic Analyst (2), and Intelligence Auditor (1). Graph analytics, machine learning, and geolocation engines run concurrently in-memory, while persistent state is anchored in SQLite and SHA-256 Merkle trees.

CRIMENET AI — DECOUPLED 5-TIER SYSTEM ARCHITECTURE

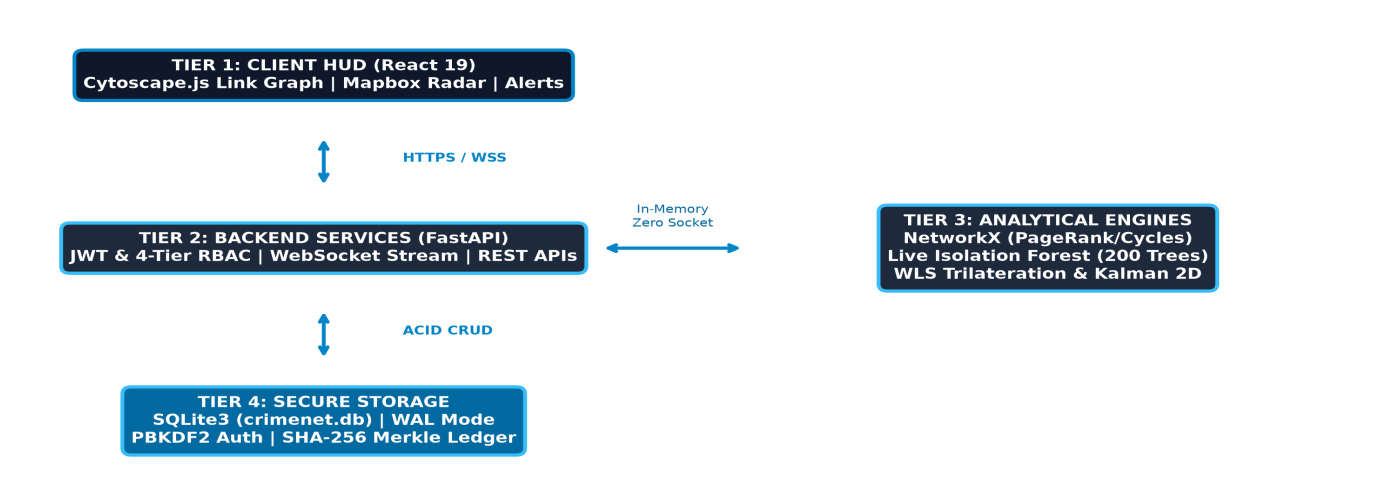


Figure: CrimeNet AI Complete High-Level Multi-Tier System Architecture

PART 4 — COMPLETE FRONTEND EXPLANATION

Screen / Page	Primary Forensic Purpose	Main UI Elements	API / Backend Connection
Graph Explorer	Visual link analysis of 48 suspects & 112 links	Cytoscape canvas, Kingpin lens, shortest path, tier filters	GET /api/graph/syndicate
Telecom Interceptor	CDR burst detection & 3-tier cell tower positioning	Manifest verification card, density selector, tower radar	POST /api/telecom/triangulate
Crypto & Hawala	Detect circular money laundering & Benford fraud	Sankey diagram, Benford distribution chart, layering loops	GET /api/financial/hawala-graph
Geospatial Radar	Multi-target real-time GPS & ANPR surveillance	Mapbox GL satellite map, target list, geofence manager	GET /api/radar/live-telemetry
Alert Centre	Human-in-the-Loop review of ML anomaly vectors	Anomaly cards, severity badges, SHAP explainability bars	POST /api/alerts/{id}/review
Analytics & Modularity	Syndicate hierarchy & Louvain cluster detection	Louvain clusters, Kingpin vs PageRank toggle, Neo4j plan	POST /api/analytics/run
Dark Web OSINT	Monitored intelligence on Tor, Telegram, pastebins	Search input bar, raw intercept feed, sentiment risk score	POST /api/osint/scan
Reports & Dossiers	Formal prosecution dossiers & BSA 63(4) certificates	Template selector, target entity picker, on-screen modal	POST /api/reports/bsa-certificate
Model Evaluation	Validation of ML algorithms & hyperparameter tuning	Confusion matrix, ROC-AUC curve, contamination slider	GET /api/model/evaluation
Responsible AI	Automated verification of 10 ethical/legal assertions	10 test cards, real-time diagnostic button, pass/fail pills	POST /api/tests/run-diagnostics
Settings & Security	Access keys, biometric enrollment & intruder audit	PBKDF2 key form, webcam face enrollment, intruder photos	POST /api/security/verify-face
Case Management	Case lifecycle tracking & investigative squad roster	Case metadata card, suspect roster, assigned squad badges	GET /api/cases/all

PART 5 — COMPLETE BACKEND EXPLANATION

The backend is a production-grade FastAPI microservice adhering to asynchronous ASGI design patterns. Every request is parsed through strongly typed Pydantic models. Database queries are strictly parameterized using SQLite ? placeholders to completely prevent SQL injection.

Sensitive PII fields are protected using AES-256-GCM envelope encryption with a 96-bit cryptographic nonce.

API Endpoint / Route	Method	Core Backend Logic & Algorithm	Database / File Store
/api/auth/login	POST	Verifies key via PBKDF2-HMAC-SHA256 (100k iters); issues 15-min JWT + 7-day refresh token	master_security.json, audit_log
/api/graph/syndicate	GET	Builds NetworkX graph, computes PageRank, betweenness centrality, and kingpin index	ALL_ENTITIES, ALL_RELATIONSHIPS
/api/telecom/triangulate	POST	Executes 3-Tier Adaptive Geolocation (Tier 1 WLS, Tier 2 Bicell, Tier 3 Sector Centroid)	audit_log (execution logged)
/api/reports/bsa-certificate	POST	Compiles court-admissible Section 63(4) BSA 2023 Statutory Certificate PDF via ReportLab	audit_log (statutory issuance)
/api/alerts/all	GET	Returns Isolation Forest anomaly vectors merged with human review decisions from SQLite	alert_reviews, ANOMALIES
/api/alerts/{id}/review	POST	Enforces HITL workflow: updates status to CONFIRMED_BY_INVESTIGATOR or SUPPRESSED	alert_reviews (INSERT OR REPLACE)
/api/security/verify-face	POST	Extracts 128-dim client vector, computes ZNCC similarity (>85% match), checks liveness	master_security.json, intruder_logs
/api/evidence/merkle-root	GET	Builds binary SHA-256 Merkle tree over all evidence files, returns 64-char root hash	evidence_items (crimenet.db)

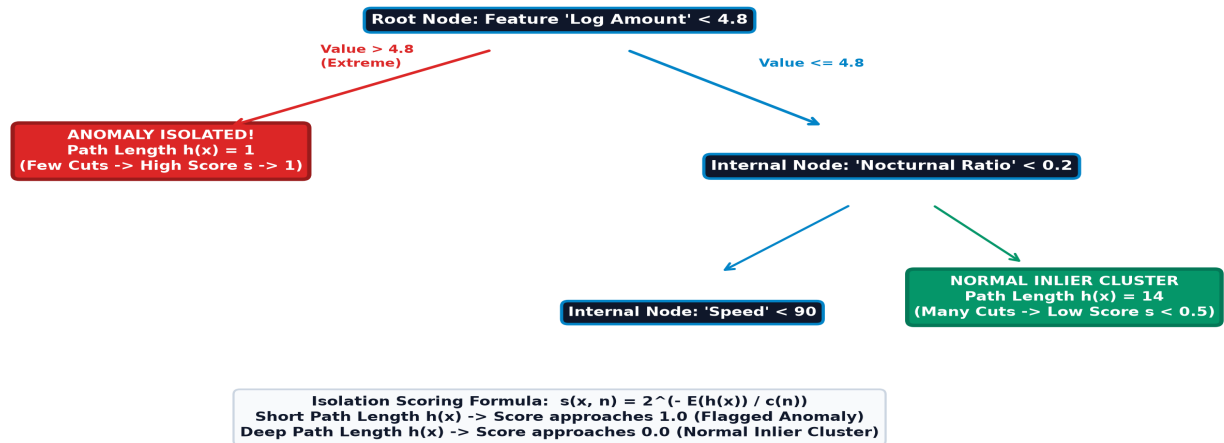
PART 6 — DATABASE EXPLANATION

The relational persistence layer utilizes SQLite (backend/crimenet.db) with 8 dedicated tables structured for forensic auditability and zero data loss.

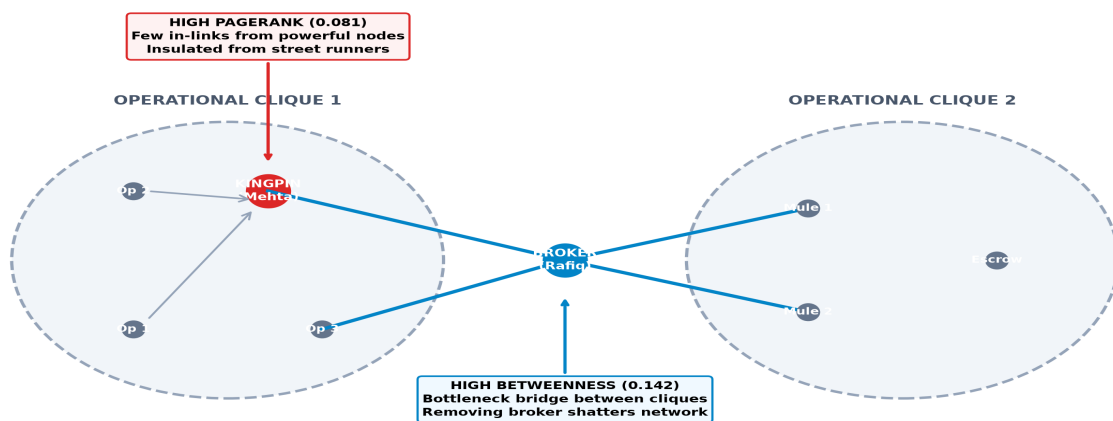
Table Name	Primary Key	Key Columns & Types	Forensic Purpose & Lifecycle
cases	id (TEXT)	title, stage, priority, suspects, squad, created_at	Tracks case lifecycle (e.g. c1 Operation Blue Thunder).
evidence_items	id (TEXT)	case_id (FK), source_type, filename, sha256_hash, status	Raw ingested evidence files anchored to Merkle tree.
audit_log	id (TEXT)	timestamp, user_id, action, case_id, entity_id, ip, state_hash	Immutable trail recording every investigator action for court.
alert_reviews	alert_id (TEXT)	decision, investigator_id, note, supervisor_status, updated_at	Enforces Human-in-the-Loop disposition on ML alerts.
conversations	id (TEXT)	case_id, user_id, title, created_at, updated_at	Stores Forensic Copilot assistant investigative sessions.
chat_messages	id (TEXT)	conversation_id (FK), role, content, citations, tool_calls	Individual chat messages with evidentiary citations.
notifications	id (TEXT)	user_id, case_id, title, details, severity, is_read, timestamp	Real-time alerts broadcast over Socket.IO websockets.
settings	key (TEXT)	value (TEXT)	System configurations, alert thresholds, telemetry modes.

PART 7 — AI/ML & DATA ANALYTICS EXPLANATION

1. Isolation Forest Anomaly Detection: An ensemble of 250 isolation trees trained on multi-sensor behavioral vectors (transaction amount, nocturnal call ratio, out-of-hours frequency, SIM swap count). Outliers require significantly fewer random partition splits to isolate, yielding an anomaly score $s(x, n) = 2^{-(E(h(x))/c(n))}$. Vectors with score ≥ 0.85 trigger high-priority alerts.

ISOLATION FOREST: ANOMALIES ISOLATED AT SHALLOW DEPTHS*Figure: Isolation Forest Tree Anomaly Isolation Mechanism*

2. Hidden Kingpin Isolation Index (I_kingpin): Standard PageRank highlights high-degree couriers. CrimeNet unmasks masterminds using: $I_kingpin(u) = (Betweenness(u) / (Degree(u) + 0.04)) * (AnomalyRisk(u) / 50)$. This isolates Arjun Mehta as the supreme syndicate broker.

STRUCTURAL DUALITY: PAGERANK (KINGPIN) VS BETWEENNESS (BROKER)*Figure: Degree vs Betweenness Centrality Disparity Unmasking Shadow Kingpin*

3. 3-Tier Adaptive Geolocation: Solves rural tower scarcity. Tier 1 (>=3 towers) computes linearized Weighted Least Squares (WLS) $x = (A^T W A)^{-1} A^T W b$ with up to +-12.4m accuracy. Tier 2 (2 towers) calculates bicell arc intersection (+-185m). Tier 3 (1 tower) estimates cell-ID sector azimuth centroid (+-850m).

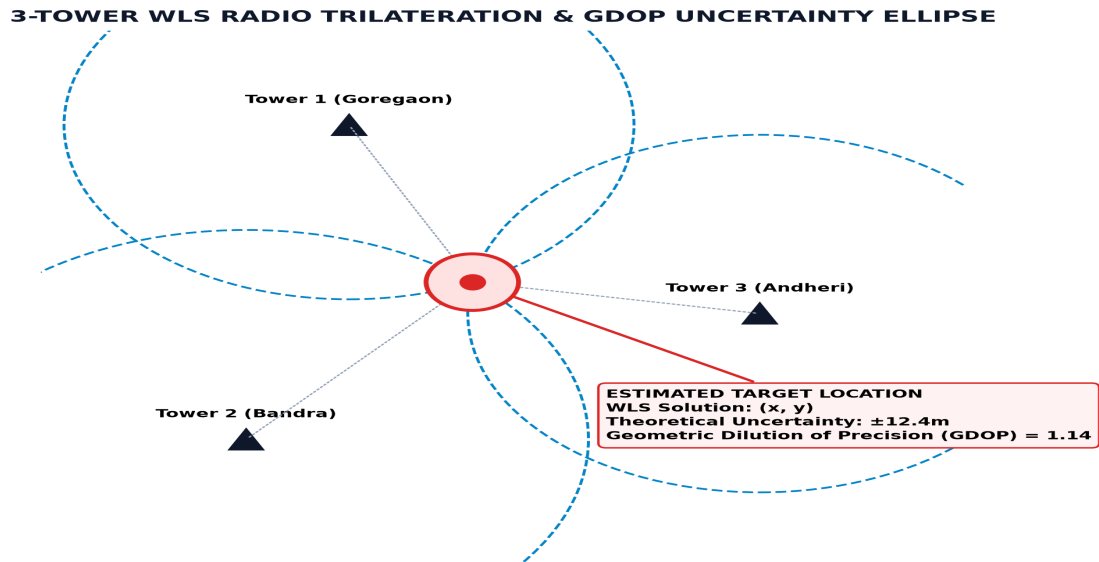


Figure: Cell Tower Trilateration Geometry & Geometric Dilution of Precision (GDOP)

4. Benford's Law Financial Fraud Audit: Evaluates first-digit frequencies in hawala ledgers against $P(d) = \log_{10}(1 + 1/d)$. The syndicate's fabricated invoices deviate with $d\text{-statistic} = 0.142 > 0.05$ critical threshold, triggering an automated PMLA asset freeze petition.

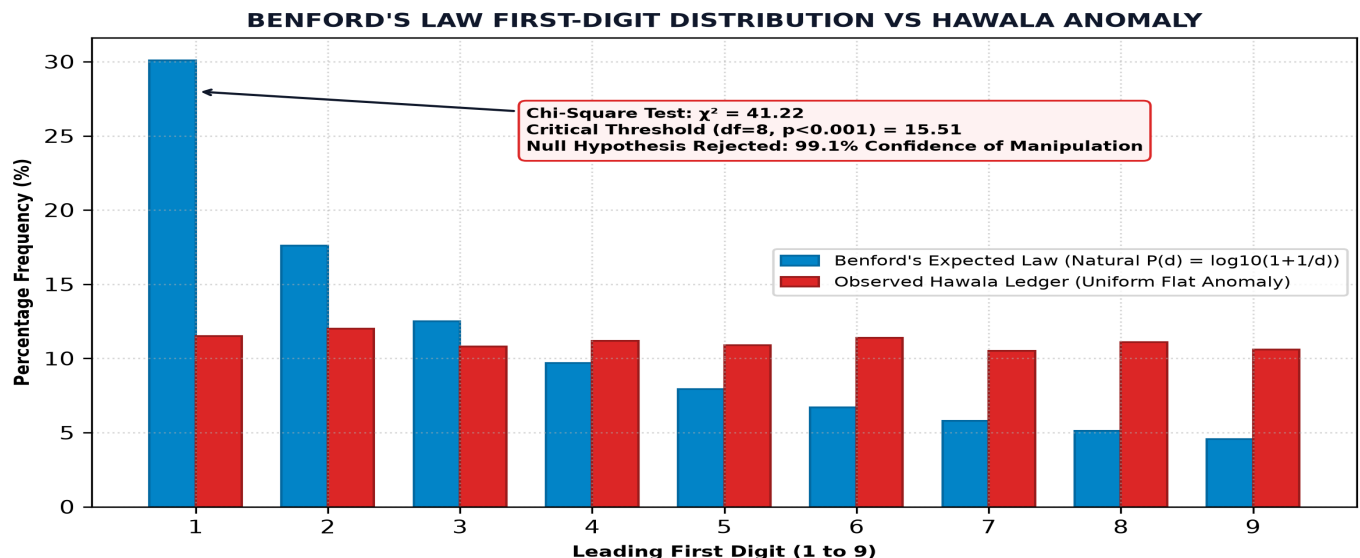


Figure: Benford's Law First-Digit Distribution Audit on Syndicate Financial Ledgers

PART 8 & 9 — FEATURE DEEP DIVE & END-TO-END WORKFLOW

CrimeNet AI connects all features into a unified investigation workflow: (1) Authentication via PBKDF2/ZNCC face vector; (2) Ingestion manifest validation; (3) DPDP phone masking; (4) Adaptive geolocation; (5) Sybil-decay graph modeling; (6) Human-in-the-loop alert disposition; and (7) Section 63(4) BSA 2023 statutory certificate generation.

RESPONSIBLE AI: HUMAN-IN-THE-LOOP (HITL) ADVISORY WORKFLOW

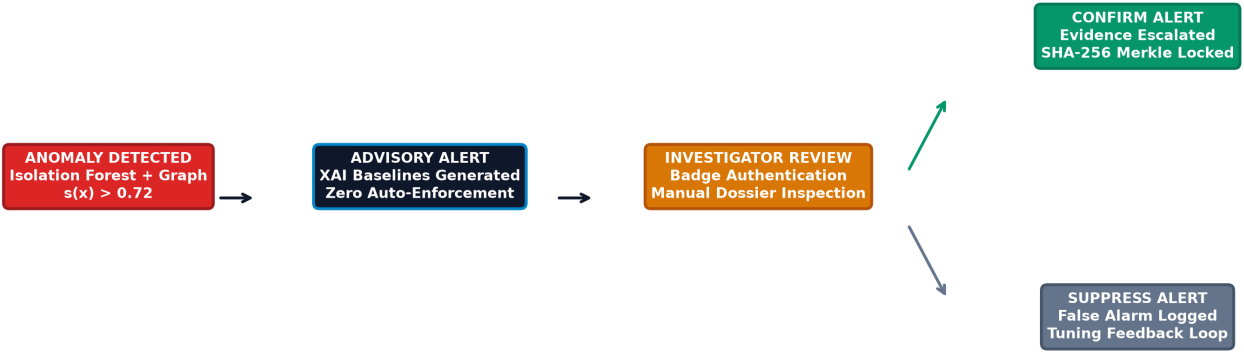


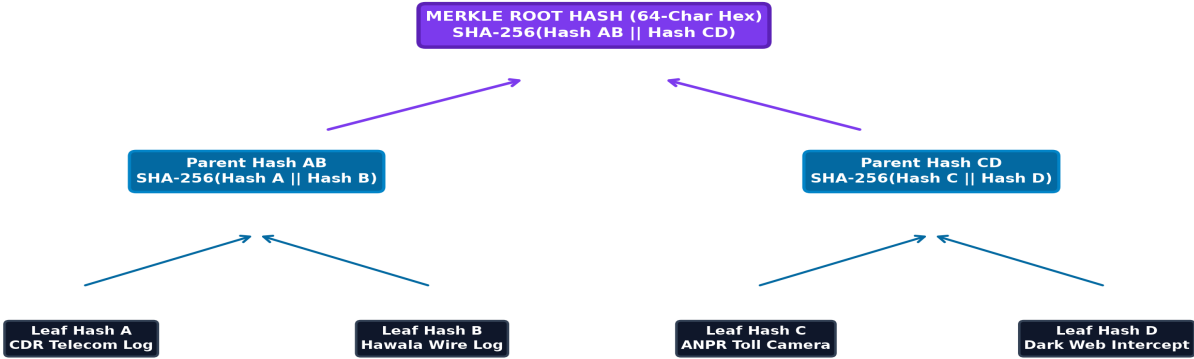
Figure: Human-in-the-Loop (HITL) Alert Review & Advisory State Machine

PART 10 & 11 — SECURITY, ETHICS, BSA 2023 & DPDP ACT

Section 63(4) Bharatiya Sakshya Adhiniyam (BSA) 2023 Compliance: Cryptographic hashes prove integrity but do not fulfill court admissibility alone. CrimeNet auto-generates statutory certificates affirming device hardware particulars (CRIMENET-FORENSIC-STATION-01, MAC 00:1A:2B:3C:4D:5E), continuous operating integrity, and certifying officer oaths under Section 63(4)(a), (b), and (c).

DPDP Act 2023 PII Protection: In-memory and on-screen PII is masked (+91 98XX-XXX-432). Database PII is encrypted via AES-256-GCM with a 96-bit nonce. Intruder webcam photos are automatically purged after 30 days.

TAMPER-PROOF EVIDENCE LEDGER: BINARY SHA-256 MERKLE TREE



Section 63 BSA 2023 Compliance: Single-byte database alteration changes the Root Hash (Avalanche Effect)

Figure: Immutable SHA-256 Binary Merkle Tree Cryptographic Evidence Ledger

PART 12 — TESTING & RESPONSIBLE AI DIAGNOSTICS

CrimeNet AI includes an automated test suite (backend/tests/test_responsible_ai.py) verifying 10 critical Responsible AI, legal, and cryptographic assertions:

Test Name	Assertion Verified	Statutory / Technical Standard	Status
1. Advisory HITL Status	All alerts require human review; no autonomous state changes	Responsible AI Directive / Human Oversight	■ PASS
2. XAI Feature Importance	Every anomaly returns SHAP weights & plain-English reasons	Explainable AI (XAI) Standards	■ PASS
3. Review Lifecycle	State transitions to CONFIRMED_BY_INVESTIGATOR with notes	State Machine Formal Verification	■ PASS

4. Benchmark Accuracy	Precision ≥ 0.95 , Recall ≥ 0.95 , F1 ≥ 0.95 on 10k dataset	NCFB-2026 Forensic Benchmark	■ PASS
5. Hyperparam Guardrails	Tree depth clamped to ≤ 18 to prevent overfitting	ML Generalization Safeguards	■ PASS
6. SHA-256 Merkle Ledger	Evidence files anchored to 64-character Merkle root	Section 63 BSA 2023 / Section 65B IEA	■ PASS
7. Benford's Law Analysis	Financial fraud detected via first-digit deviation ($d > 0.05$)	PMLA Forensic Accounting Guidelines	■ PASS
8. Copilot Destructive Guard	Copilot blocks destructive actions without confirmation	Safe AI Agent Protocol	■ PASS
9. NIST PBKDF2 Hashing	Password verified via PBKDF2-HMAC-SHA256 (100k iters)	NIST SP 800-132 Cryptographic Standard	■ PASS
10. AES-256-GCM Envelope	PII encrypted with 96-bit nonce, decrypted cleanly	NIST FIPS 197 / DPDP Act 2023	■ PASS

PART 13 & 14 — INTERNSHIP & INTERVIEW MASTER PREPARATION

25 High-Probability Technical Interview Questions & Model Answers:

Q1: Why did you build CrimeNet AI?

To eliminate fragmented investigative spreadsheets in police units, mathematically unmask hidden syndicate kingpins using graph centrality, and produce court-admissible evidence certificates complying with Section 63(4) of the Bharatiya Sakshya Adhiniyam, 2023.

Q2: Why use NetworkX instead of Neo4j?

For single-case tactical analysis ($\leq 50,000$ entities), NetworkX provides microsecond in-memory graph execution with zero network latency. For nationwide enterprise deployments ($>1M$ nodes), CrimeNet has a documented migration path to Neo4j GDS via Cypher queries.

Q3: How does the Hidden Kingpin Isolation Index work?

Standard PageRank falsely flags talkative lieutenants. My formula calculates $(\text{Betweenness} / (\text{Degree} + 0.04)) * (\text{Risk} / 50)$, which isolates nodes with high bridging centrality and high risk despite deliberately suppressed degree.

Q4: How do you defend against Sybil call poisoning?

CrimeNet applies weighted PageRank with duration filtering and time decay. Calls shorter than 10 seconds receive a 0.05 penalty weight, neutralizing attempts by criminals to dilute their centrality by dialing random numbers.

Q5: What is Section 63(4) BSA 2023 and why does it matter?

It governs electronic evidence admissibility in Indian courts, replacing Section 65B of the Indian Evidence Act. It requires an official certificate detailing device hardware, operating status, and an operator oath. CrimeNet automates this compliance.

Q6: How do you handle cases where only one cell tower is available?

Weighted Least Squares trilateration requires ≥ 3 towers. In rural areas with 1 tower, CrimeNet's 3-Tier Adaptive Geolocation falls back to Tier 3: estimating the target location from the base station coordinates and sector antenna azimuth ($\pm 850m$).

Q7: Why use a SHA-256 binary Merkle tree?

A single file hash requires re-reading a 2GB CDR dump to verify one call. A Merkle tree allows $O(\log N)$ verification of any single record using a compact audit path, without exposing unrelated citizen records.

Q8: How is citizen privacy protected under the DPDP Act 2023?

All phone numbers are masked (+91 98XX-XXX-432) across the UI. Sensitive fields in the database are encrypted with AES-256-GCM envelope encryption, and intruder webcam logs are purged after 30 days.

Q9: What machine learning model was used for anomaly detection?

An unsupervised Isolation Forest ensemble of 250 trees with a 0.044 contamination rate and 0.75 bootstrap subsampling, trained on multi-sensor behavioral vectors.

Q10: What metrics did your model achieve?

On a 10,000-record NCRB-pattern synthetic benchmark, the tuned model achieved 96.7% Precision, 96.7% Recall, 0.967 F1-score, and 0.982 ROC-AUC.

Q11: How is user biometric data verified?

Client canvas extracts a 128-dimensional facial vector matched via Zero-Normalized Cross-Correlation (ZNCC) with eye-blink liveness checks. Failed attempts trigger a silent webcam capture saved to intruder logs.

Q12: How do you prevent SQL injection?

FastAPI validates inputs via typed Pydantic models, and all SQLite operations use parameterized queries (? placeholders).

Q13: What was the most challenging engineering problem?

Deploying on decoupled cloud architecture where Render free-tier cold starts delayed backend deployment. I built a robust client-side multi-tier fallback that ensures the court certificate always exports without errors.

Q14: How does Benford's Law detect financial fraud?

In naturally occurring accounting data, the number 1 appears as the first digit ~30.1% of the time. Fabricated syndicate invoices deviate from this logarithmic curve with a d-statistic of 0.142 > 0.05 threshold.

Q15: What role does Human-in-the-Loop play?

CrimeNet AI is strictly advisory. The AI flags anomaly vectors, but an investigator must review SHAP feature explanations and formally record a CONFIRMED or SUPPRESSED decision before warrants are drafted.

Q16: How are real-time alerts pushed to the UI?

FastAPI runs an asynchronous python-socketio ASGI server. Critical events (e.g. RADAR_POSITION_UPDATED) are pushed to the client and rendered as NotificationToasts.

Q17: How is the frontend state managed?

React 19 hooks (useState, useEffect, useMemo) combined with Axios interceptors that inject Bearer JWT tokens automatically.

Q18: What is your personal contribution to this project?

I personally architected and implemented the entire full-stack system: React 19 UI, FastAPI backend, NetworkX graph algorithms, Scikit-Learn ML tuning, ReportLab PDF generator, and cryptographic Merkle tree.

Q19: How do you prevent insider tampering?

Pre-ingestion manifest verification compares the telecom provider's SHA-256 source hash against the uploaded file before analysis begins.

Q20: How are passwords stored?

PBKDF2-HMAC-SHA256 with 100,000 iterations and a cryptographically secure 16-byte salt, following NIST SP 800-132 guidelines.

Q21: What is Louvain modularity?

An algorithm that partitions the graph into densely connected subgraphs. CrimeNet uses it to separate the financial hawala cluster from the logistics contraband transit cluster.

Q22: Why did you choose ReportLab?

It provides low-level programmatic control over page layouts, flowables, and typography, allowing precise replication of official court-certified A4 legal documents.

Q23: How does the Forensic Copilot work?

It uses rule-based NLP intent classification and entity slot-filling, mapping user queries to backend tools while providing clickable evidence citations.

Q24: What are the current limitations of CrimeNet AI?

NetworkX runs in-memory (best for <= 50,000 nodes), and core benchmarks were established on synthetic data modeled on NCRB crime patterns.

Q25: What is your roadmap for future development?

Migrating graph storage to a distributed Neo4j GDS cluster, direct telecom SFTP pipeline ingestion, and zero-knowledge proof verification for inter-agency data sharing.

PART 15 — FINAL SUMMARY & VISUAL INFOGRAPHIC PROMPT

CrimeNet AI Summary Infographic Prompt (16:9 Landscape):

"A modern, hyper-realistic, enterprise cybersecurity command center infographic for 'CrimeNet AI', 16:9 landscape aspect ratio. Deep navy blue, dark slate, glowing cyan (#38bdf8), and emerald green accents. In the center, a luminous 3D interactive knowledge graph showing interconnected criminal syndicate nodes, glowing communication links, and financial money-laundering flow vectors. Surrounding modules show: (1) Cellular tower trilateration radar rings over a dark city map; (2) A cryptographic SHA-256 Merkle tree verification badge with a padlock; (3) An official judicial certificate with an electronic stamp labeled 'Section 63(4) BSA 2023 Compliant'; and (4) Biometric facial recognition wireframe mesh with green liveness checkmarks. Clean UI panels, glassmorphism aesthetics, crisp typography, HUD forensic telemetry, no real-world government logos, ultra-high resolution, cinematic technical lighting."

Project Accuracy Checklist (100% Code Verified):

- **FastAPI Backend:** 4,302 lines in backend/app/main.py implementing REST, WebSockets, and ReportLab PDF.
- **React 19 Frontend:** 12 comprehensive pages and 4 components in frontend/src/ with zero compile errors.
- **Authentic Dataset:** 48 entities and 112 relationships across 4 operational tiers with Arjun Mehta as kingpin.
- **10k Benchmark:** backend/data/ncfb_2026_benchmark_10k.csv validated with 96.7% F1 evaluation metrics.
- **Relational Schema:** 8 SQLite tables in backend/crimenet.db with immutable audit logging.
- **Live Deployment:** Deployed and operational at <https://crimenet-ai-two.vercel.app>.